

8 OPTIONAL FUNCTIONS

8.1 Summary

This chapter describes the optional functions set for this device.

<Contents>

- Count set weighing
- Parent and child weighing

<Purpose>

- To understand and master the optional functions.

<Intended reader>

- System administrators

8.2 Parent and Child Weighing

Parent and child weighing is a weighing method that performs mix weighing by discharging the product of a certain weigh head (parent head) without fail.

TIP

- Heads other than the parent head are referred to as "child head".
When performing the parent and child weighing, mechanical measures must be taken to prevent the product weighed by the parent head from mixing with products weighed by child heads.

8.2.1 Types of Parent and Child Weighing

The parent and child weighing is a weighing method that performs mix weighing based on the parent head weighing results, while the standard mix weighing is based on the combination results of each weighed product.

There are two types of parent and child weighing: "Parent and child mix weighing" and "Bonus adding".

8.2.1.1 Parent and Child Mix Weighing

"Parent and child mix weighing" is a weighing method in which the parent head weight is included into the target weight (the parent head participates in the combination weighing).

The device first weighs the product in the parent head. The target weight for child heads is calculated based on the parent head weight.

Target weight for child heads = Target weight - Parent head weight

Example: When the target weight is 200.0 g and the parent head weight is 30.5 g, the target weight for child heads will be:

$$200.0 - 30.5 = 169.5 \text{ (g)}$$

The combination weighing will be performed with the child head target weight set to 169.5 g.

The discharged product weight will be the sum of the combination weighing result with the child head target weight of 169.5 g, and the parent head weight.

For instance, when the combination weighing result of the child head is 170.5 g, the discharged product weight will be $170.5 + 30.5 = 201.0 \text{ (g)}$.

8.2.1.2 Bonus Adding

"Bonus adding" is a weighing method in which the parent head weight is not included into the target weight (the parent head does not participate in the combination weighing).

The target weight for child heads will be the target weight of the device.

Target weight for child heads = Target weight

Example: When the target weight is 200.0 g and the parent head weight is 30.5 g:

$$200.0 + 30.5 = 230.5 \text{ (g)}$$

The discharged product weight will be the sum of the combination weighing result with the child head target weight of 200.0 g, and the parent head weight.

For instance, when the combination weighing result of the child head is 200.5 g, the discharged product weight will be

$$200.5 + 30.5 = 231.0 \text{ (g)}$$

8.2.2 Setting Items of Parent and Child Weighing

The Preset screen is used to configure the parent and child weighing settings. Setting items for the parent and child weighing are added to the Preset screen.

Refer to the following figure and table for setting items.



Fig. 8-1

Table 8-1

No.	Name	Function
1	[Parent and Child] radio button	Selects [On] or [Off] for the parent and child weighing. [On]: Activates the parent and child weighing. [Off]: The device performs normal weighing.
2	[Parent Head Count] radio button	Sets "1" or "2" as the number of parent head(s).
3	[Parent Hd. Participation] radio button	Selects [On] or [Off] for the participation of the parent head in the combination weighing. [On]: Switches into the parent and child mix weighing mode. [Off]: Switches into the bonus adding mode.

Table 8-1

No.	Name	Function
4	[Parent Zero Adjust] key	Sets the interval to perform zero adjustment for the parent head. 0:Zero adjustment operation is not performed for the parent head. Other than 0:Interval for the zero adjustment is set. The number of times that the parent head discharges products is counted. For example, when "100" is set as the interval, zero adjustment is performed in every 100 discharges from the parent head. NOTE When the automatic zero adjustment interval is set to "0" in the weigher specification setting screen, the zero adjustment interval for the parent head cannot be specified.
5	[Parent Dump Mode] radio button	When the number of parent heads is set to "2", selects discharging mode from [Alternate] or [Simul.]. [Alternate]:Products are discharged first from P1 parent head and then from P2 parent head. [Simul.]:Each parent head discharges products simultaneously.
6	[Action Over Upper Limit] radio button	Selects the action when the parent head weight exceeds the upper weight limit that is described later in this section. [PrntDump]:Discharges products from the parent head only. [P & C Dump]:Discharges products from both parent head and child heads. NOTE If the parent head weight reaches the full scale, the alarm buzzer sounds and the discharge operation is not performed.
7	[Action Under Lower Limit] radio button	Selects the action when the parent head weight falls below the lower weight limit that is described later in this section. [PrntDump]:Discharges products from the parent head only. [Add'n'l Feed]:Does not discharge products and supplies additional products. NOTE If the parent head weight is judged as an under scale, the alarm buzzer sounds without supplying additional products.
8	[Parent Head Setting] index	Displays the Parent Head Setting screen.

When the parent head count is set to "2", the following items must be set to each parent head respectively.
When the parent head count is changed to "2" and the parent head setting is not yet completed, the parent head setting index blinks to prompt you to complete the setting.

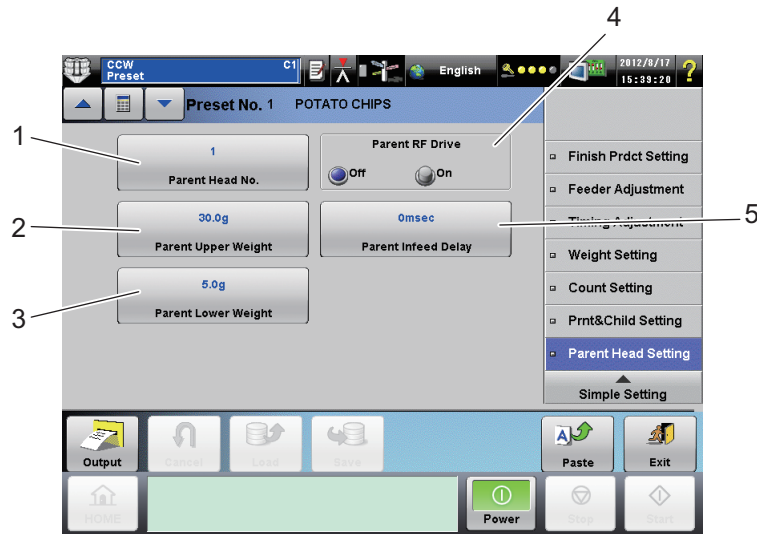


Fig. 8-2

Table 8-2

No.	Name	Function
1	[Parent Head No.] key	Sets the weigh head number of the parent head. NOTE Specify the weigh head number that is included in the section. If the weigh head number of a different section is specified, an error message will appear.
2	[Parent Upper Weight] key	Sets the upper weight limit for the parent head. Products are discharged when the parent head weight is within the specified range. NOTE If the specified value is smaller than the value set for the parent lower weight limit, both the parent upper weight key and the parent lower weight key blink to prompt you to modify the setting.
3	[Parent Lower Weight] key	Sets the lower weight limit for the parent head. Products are discharged when the parent head weight is within the specified range. NOTE If the specified value is larger than the value set for the parent upper weight limit, both the parent upper weight key and the parent lower weight key blink to prompt you to modify the setting.

Table 8-2

No.	Name	Function
4	[Parent RF Drive] radio button	Selects [On] or [Off] for activation of the parent head radial feeder. [On]: Activates the parent head radial feeder. [Off]: Does not activate the parent head radial feeder. NOTE The parent head radial feeder is not automatically controlled.
5	[Parent Infeed Delay] key	Specifies the time from when the parent head pool hopper opens until when the feed signal is output to the external device. Setting range: 0 to 2550 msec (in 10 msec)

8.2.3 Setting Procedures of Parent and Child Weighing

For setting the parent and child weighing, follow the procedures below.

- On the Main Menu, press the [Preset] key.

► The Preset screen appears.



Fig. 8-3

- Press the [Prnt&Child Setting] key in the index.

► The Prnt & Child Setting screen appears.

TIP

- Delay amount of the parent head can be specified via the Timing Adjustment screen in the index.
- WH DELAY amount can be specified via the Timing Adjustment screen in the index.



Fig. 8-4

3. Press the [Parent and Child] radio button and select [On].
4. Press the [Parent Head Count] radio button and select [1] or [2].
5. Press the [Parent Hd. Participation] radio button and select [On] or [Off].
6. Press the [Parent Zero Adjust] key, and input the numerical value with the Numeric Keypad screen displayed on the screen.
7. Press the [Parent Dump Mode] radio button and select [Alternate] or [Simul.].
8. Press the [Action Over Upper Limit] radio button and select [PrntDump] or [P & C Dump].
9. Press the [Action Under Lower Limit] radio button and select [PrntDump] or [Add'nl Feed].
10. Press the [Parent Head Setting] key in the index.

► The Parent Head Setting screen appears.

TIP

- When the number of parent heads is set to [2], the [Parent Head Selection] drop-down key appears.

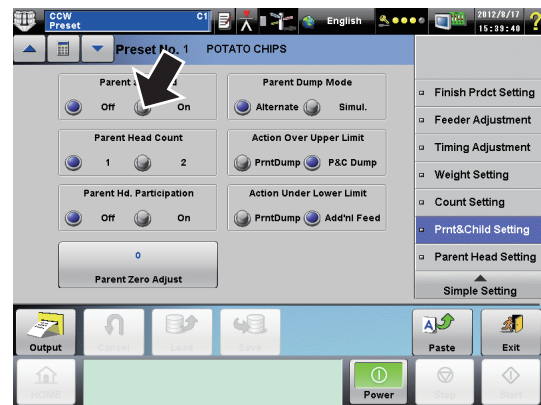


Fig. 8-5



Fig. 8-6

11. Press the [Parent Head No.] key, and input the numerical value with the Numeric Keypad screen displayed on the screen.
12. Press the [Parent Upper Weight] key, and input the numerical value with the Numeric Keypad screen displayed on the screen.
13. Press the [Parent Lower Weight] key, and input the numerical value with the Numeric Keypad screen displayed on the screen.
14. Press the [Parent RF Drive] radio button and select [On] or [Off].
15. Press the [Parent Infeed Delay] key, and input the numerical value with the Numeric Keypad screen displayed on the screen.
16. Press the [Exit] key.

► The display returns to the Main Menu.



Fig. 8-7

8.2.4 Production screen of Parent and Child Weighing

On the Production screen for the parent and child weighing, the parent head weight is displayed on the Combination tab screen as shown in the figure.

<Production screen during production with the parent and child weighing>

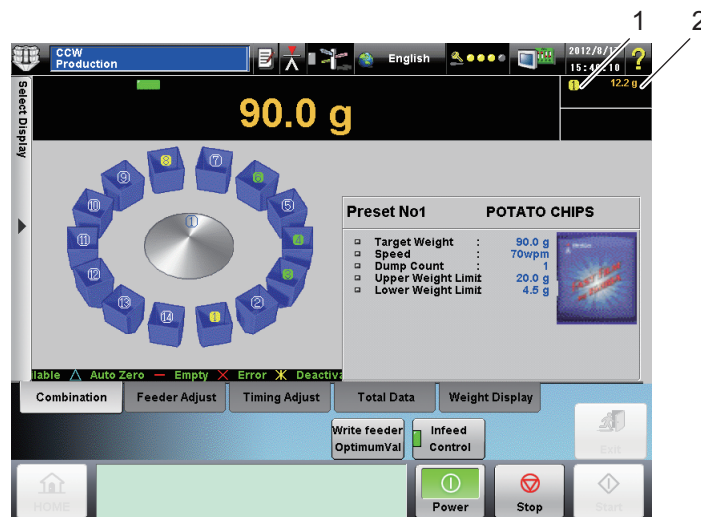


Fig. 8-8

Table 8-3

No.	Name
1	Parent head number
2	Parent head weight

8.3 Count Set Weighing

Count set weighing is a method in which the number of pieces of product is set as the target value, while the product weight is set as the target in normal weighing. The weight of the product supplied to each weigh hopper is divided by the mean piece weight and converted to the number of pieces for combination.

8.3.1 Setting Items of Count Set Weighing

The Count Setting index screen of the Preset screen is used to configure the count set weighing settings. Setting items for the count set weighing are added to the Preset screen.

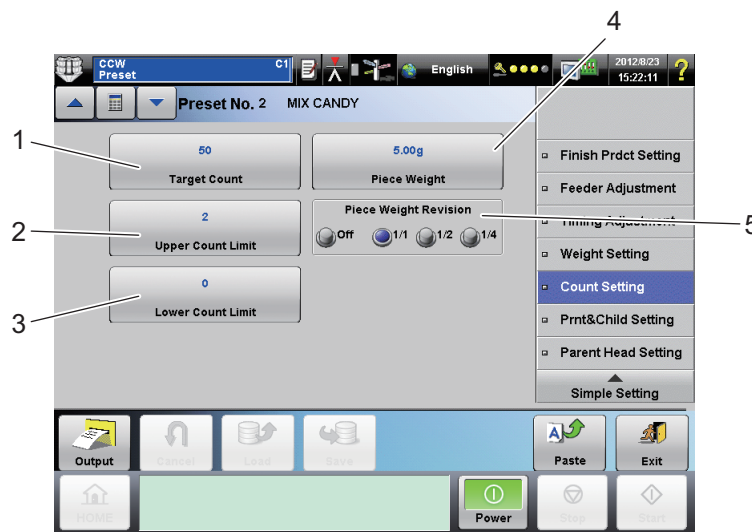


Fig. 8-9

Table 8-4

No.	Name	Function
1	[Target Count] key	Sets the target number of pieces. NOTE When not performing the count set weighing, set to "0".
2	[Upper Count Limit] key	Sets the upper count limit which is the number of permissible pieces exceeding the target value. When the upper count limit is set to "0", weighing results that exceed the target count are not accepted.
3	[Lower Count Limit] key	Sets the lower count limit which is the number of permissible pieces below the target value. When the lower count limit is set to "0", weighing results that fall below the target count are not accepted.
4	[Piece Weight] key	Sets the weight for a single piece of product. The number of pieces supplied to each hopper is calculated based on this piece weight value.

Table 8-4

No.	Name	Function
5	[Piece Weight Revision] radio button	Specifies whether the device automatically updates the piece weight ([1/1], [1/2], [1/4]) or not ([Off]). When set to [Off], the number of pieces supplied to each weigh hopper is calculated based on the specified piece weight value. When the piece weight varies, set this field to perform update so that the piece weight is automatically updated and the number of pieces supplied to each weigh hopper is calculated based on the updated piece weight value. Furthermore, there are three options for the update setting ([1/1], [1/2], and [1/4]) depending on the range to which the updated operation can be performed. Set this option to [1/1] in normal operation. [1/1]: Use this option if no error occurs in converting to the number of pieces when the target amount of product is supplied to one head. [1/2]: Use this option if no error occurs in converting to the number of pieces when half of the target amount of product is supplied to one head. [1/4]: Use this option if no error occurs in converting to the number of pieces when one quarter of the target amount of product is supplied to one head. TIP Refer to "8.3.2 Piece Weight Revision" for details regarding the piece weight update.

NOTE

- In count set weighing, after the combination weighing is completed, the weighing result is checked to see if it is within "target weight + upper weight limit" and "target weight-lower weight limit".

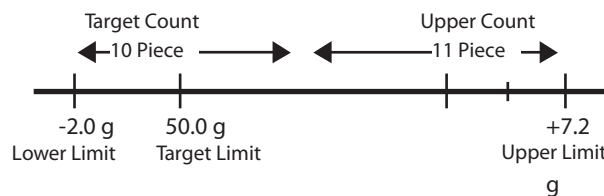
Products will not be discharged if the value is not within this range, so care must be taken in setting upper and lower weight limits. A setting example is shown below.

Example: When the target count is 10, the upper count limit is 1, and the piece weight is 5.00 g (4.80 to 5.20 g)

Target weight: 50 g (5.00×10)

Lower weight limit: 2 g ($50.0 - 4.80 \times 10$)

Upper weight limit: 7.2 g ($5.20 \times 11 - 50.0$)



TIP**Auto settings of [Target Count]/[Upper Count Limit]/[Lower Count Limit]**

- When entering the values for the target count, upper count limit, lower count limit, piece weight and piece weight revision, the target weight, upper weight limit and lower weight limit values are automatically set as follows (Values other than "0" should be specified for the target count and piece weight).

Target weight = Target count × Piece weight

Upper weight limit = Upper count limit × Piece weight + Piece weight/2 (When the piece weight revision is set to [1/1])

= Upper count limit × Piece weight + Piece weight (When the piece weight revision is set to [1/2])

= Upper count limit × Piece weight + 2 × Piece weight (When the piece weight revision is set to [1/4])

= Upper count limit × Piece weight + 2 × Piece weight (When the piece weight revision is set to [Off])

Lower weight limit = Lower count limit × Piece weight + Piece weight/2 (When the piece weight revision is set to [1/1])

= Lower count limit × Piece weight + Piece weight (When the piece weight revision is set to [1/2])

= Lower count limit × Piece weight + 2 × Piece weight (When the piece weight revision is set to [1/4])

= Lower count limit × Piece weight + 2 × Piece weight (When the piece weight revision is set to [Off])

Upper weight limit and lower weight limit values set by the auto calculation are not rechecked when the piece weight varies within the range where conversion to the number of pieces is performed without error.

8.3.2 Piece Weight Revision

When this function is activated, the piece weight is updated within the range where the device can recognize the number of pieces of product supplied to each weigh hopper automatically until the product amount reaches the target count.

This range is called the piece weight revision range.

Usually, the piece weight revision range can be calculated as follows, where:

TC:Target count

PW:Piece weight

PWRev:Piece weight revision (1/1, 1/2, 1/4)

$$PW (1 - (1/(2TC \times PWRev))) < \text{Piece weight revision range} < PW (1 + (1/(2TC \times PWRev)))$$

For example, the piece weight revision range can be calculated as follows, where:

Target count value:20

Piece weight:5 g

Piece weight revision:1/1

$$5 (1 - (1/(2 \times 20 \times 1/1))) < \text{Piece weight revision range} < 5 (1 + (1/(2 \times 20 \times 1/1)))$$

$$4.875 \text{ g} < \text{Piece weight revision range} < 5.125 \text{ g}$$

In this example, update operation will be automatically performed in every 5 cycles with the piece weight range between 4.875 g and 5.125 g. Also the number of pieces of product supplied into each hopper can be identified within the range between 1 and 20 (target count value).

If the updated piece weight exceeds the piece weight revision range, [PIECE WEIGHT ERROR] is displayed and the device stops the production.

Each piece weight revision range in this example is shown in "Table 8-5".

Table 8-5

Setting Value for Piece Weight Revision	Piece Weight Revision Range (Target count:20, piece weight:5 g)	Recognition Range
1/1	4.875 g < Piece weight revision range < 5.125 g	1 to 20
1/2	4.75 g < Piece weight revision range < 5.25 g	1 to 10
1/4	4.5 g < Piece weight revision range < 5.5 g	1 to 5

When the value for the piece weight revision is set to 1/1, the piece weight revision range is more restricted, but more products supplied into each hopper can be identified (maximum target count value). When set to 1/4, the piece weight revision range broadens, but the amount of products supplied into each hopper and identified by the device is more restricted.

8.3.3 Setting Procedures of Count Set Weighing

For setting the count set weighing, follow the procedures below.

1. On the Main Menu, press the [Preset] key.

► The Preset screen appears.



Fig. 8-10

2. Press the [Count Setting] key in the index.

► The Count Setting index screen appears.



Fig. 8-11

3. Press the [Target Count] key, and input the numerical value with the Numeric Keypad screen displayed on the screen.
4. Press the [Upper Count Limit] key, and input the numerical value with the Numeric Keypad screen displayed on the screen.
5. Press the [Lower Count Limit] key, and input the numerical value with the Numeric Keypad screen displayed on the screen.
6. Press the [Piece Weight] key, and input the numerical value with the Numeric Keypad screen displayed on the screen.
7. Press the [Piece Weight Revision] radio button and select [Off], [1/1], [1/2], or [1/4].
 - ▶ Set this option to [1/1] in normal operation.
8. Press the [Exit] key.
 - ▶ The display returns to the Main Menu.

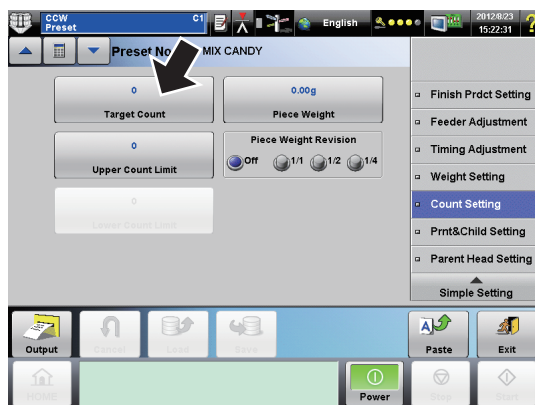


Fig. 8-12

8.3.4 Select Display Pop-up Menu (Count Set Weighing)

The count display of the Combination Result Display screen and the Piece Weight Display screen are added to the Combination tab screen for the count set weighing.



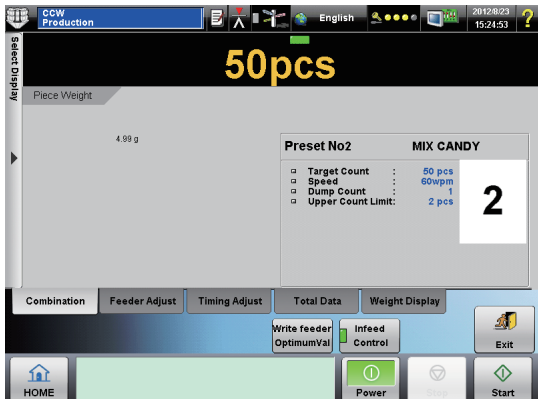
Fig. 8-13

Table 8-6

No.	Name	Function
1	[Combination Display] key (Weight display)	Selects the weight display of the Combination Result Display screen.

Fig. 8-14

Table 8-6

No.	Name	Function
2	[Combination Display] key (Count display)	Selects the count display of the Combination Result Display screen.  Fig. 8-15
3	[Piece Weight] key	Selects the Piece Weight screen.  Fig. 8-16
4	[Expansion Display] key	Selects Expansion Display of the Combination screen.  Fig. 8-17

8.3.5 Output During Count Set Production

During the count set weighing, output data display mode is changed from the weight display to the count display as shown in the following.

	TOTAL DATA	:	6	
	PRESET NUMBER:		6	
	DATE	:	2006/AUG/ 9	
	START TIME		17:36	
	STOP TIME		17:42	
1	---	C1 TOTAL	---	
	PROD CODE	:		
2	0003			
	PROD NAME	:		
3	CANDY			
	PROPER	:	119	
4	TARGET CNT:		50 pcs	
5	TOTAL	:	5964 pcs	
6	MEAN	:	50.1 pcs	
7	STANDARD D:		0.372 pcs	
8	MAX.	:	55 pcs	
9	MIN.	:	50 pcs	
	RANGE	:	5 pcs	
10	PW MEAN	:	4.97 g	
11	STANDARD D	:	0.0 g	
12	PW MAX.	:	5.10 g	
13	PW MIN.	:	4.88 g	
	78 0		0	
	77 0		0	
	76 0		0	
	75 0		0	
	74 0		0	
	73 0		0	
	72 0		0	
	71 0		0	
	70 0		0	
	69 0		0	
	68 0		0	
	67 0		0	
	66 0		0	
	65 0		0	
	64 0		0	
	63 0		0	
	62 0		0	
	61 0		0	
	60 0		0	
	(Continues to the right side)			
	(Continued from the left side)			
	59 0		0	
	58 0		0	
	57 0		0	
	56 0		0	
	55 0		0	
	54 0		0	
	53 0		0	
	52 0		0	
	51 0		0	
	50 *****9		107	
	49 0		0	
	48 0		0	
	47 0		0	
	NO OF * = 18			
14	UNDER	:	2	
	OVER	:	0	
	OVER SCALE	:	0	
	RECHECK CNT:		0	
	OVW DMP CNT:		0	
	OVS DMP CNT:		0	
	TOTAL	:	0 kg	
	METAL DETECT:		0	
	ON TIME		0:01	
	OFF TIME		0:00	
	LOW TIME		0:00	

Fig. 8-18

Table 8-7

No.	Name	Remarks
1	Product code	—
2	Product name	—
3	Proper weight	Number of times that the proper discharge is performed.
4	Target weight	—
5	Total value	Total discharged weight.
6	Mean value	—

Table 8-7

No.	Name	Remarks
7	Standard deviation	—
8	Maximum weight	—
9	Minimum weight	—
10	Weight range	(Maximum value)-(Minimum value)
11	Average piece weight	—
12	Maximum piece weight	—
13	Minimum piece weight	—
14	Quantity per * mark	Weighing results are displayed in order of weight value, histogram, and number of weighing times from the left. The number displayed in the histogram indicates a percentage of * (e.g. number $\times 10\%$). If a value of * = 18, when the number of weighing times is 45, it will be displayed as “**5” ($18 \div 18 + 18 \times 50\%$).